

SENTIMENT AI CHATBOT

Project Report

Submitted By:
Prathwik H Devadiga

1. Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone of text. This project develops an AI chatbot capable of identifying whether a user's message is Positive, Negative, or Neutral using Machine Learning.

2. Objective

- Build an AI chatbot
- Predict user sentiment
- Generate suitable responses
- Save conversation history
- Display chat summary

3. Technologies Used

- Python

- Scikit-learn
- Pandas
- Joblib
- VS Code

4. Machine Learning Algorithm

- **Algorithm:** Multinomial Naive Bayes
- **Vectorizer:** TF-IDF Vectorizer

5. Dataset

- **Dataset file:** `dataset/sentiment_dataset.csv`
- **Columns:** `text` , `sentiment`

6. Project Workflow

1. Load dataset
2. Preprocess text
3. Convert text using TF-IDF
4. Train Naive Bayes model
5. Save model and vectorizer
6. Load model
7. Predict sentiment
8. Generate chatbot response
9. Save chat history

7. Folder Structure

```
Sentiment-Analysis-Chatbot
|
├── chatbot.py
├── train_model.py
├── test_model.py
├── dataset
├── models
├── notebook
├── images
├── report
└── chat_history
```

8. Results

The chatbot successfully predicts:

- Positive 😊
- Negative 😞
- Neutral 😐

It also stores chat history and displays a summary after the conversation ends.

9. Advantages

- Easy to use
- Fast sentiment prediction
- Machine Learning based
- Stores conversations

- Simple interface

10. Applications

- Customer support
- Feedback analysis
- Social media monitoring
- Mental wellness assistance
- Product review analysis

11. Future Enhancements

- Voice chatbot
- GUI using Tkinter
- Web application
- Deep Learning models
- Multi-language support

12. Conclusion

The Sentiment AI Chatbot successfully detects user sentiment using Machine Learning. It provides suitable responses, maintains conversation history, and demonstrates the practical use of NLP techniques in chatbot applications.